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Device and method for sensing an over-temperature of a power semiconductor

Abstract

The present invention concerns a device and a method for sensing an over-temperature of a power semiconductor. The invention: provides a current pulse source through control electrodes of the power semiconductor, duplicates the current provided by the current pulse source and provides the duplicated current to an emulating device, compares the voltage across the control electrodes to the voltage across the emulating device, notifies the result of the comparison.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to a device and a method for sensing an over-temperature of a power semiconductor.

BACKGROUND ART

(2) Power semiconductor devices are made to work below a maximum junction temperature. The operation over the maximum junction temperature may trigger a thermal runaway condition that can lead to an irreversible and catastrophic failure of the power semiconductor device. The operation over the maximum junction temperature may be caused by different situations such as a defective cooling system, transient operation of the power semiconductor over its specifications, ageing and others.

(3) Nowadays, power semiconductor devices like transistors with embedded diode used as a sensor for the chip temperature are used. The diode forward voltage drop is dependent on the temperature. This solution leads to a loss of the transistor available active area and increases the number of connections to the power semiconductor.

(4) Direct measurements of the junction temperature using thermocouples or negative temperature coefficient resistors are not compatible to detect the over-temperature, given the low bandwidth of the sensors and the fact that they can't be placed close to the power semiconductor.

(5) Optical measurements of the temperature are also slow since they require digital treatment and calibration which increase the latency.

(6) The use of thermally sensitive electrical parameters of the power semiconductor is promising. For example the temperature of the power semiconductor may be observed by the injection of a measurement current into the active power semiconductor to measure the on-state voltage that is dependent of the temperature.

(7) Consequently, as this measurement needs to occur in a known state of conducted current or calibrated for all conducted currents, this increases the complexity of the measurement circuit and furthermore the precision associated to this measurement may not be enough and thus may cause false triggers.

SUMMARY OF INVENTION

(8) The present invention aims to provide a low complexity and accurate device and method for sensing an over-temperature of a power semiconductor.

(9) To that end, the present invention concerns a device for sensing an over-temperature of a power semiconductor, characterized in that the device comprises: a current pulse source that is provided through control electrodes of the power semiconductor, a current copier that duplicates the current provided by the current pulse source, and that provides the duplicated current to an emulating device, a comparator that compares the voltage across the control electrodes to the voltage across the emulating device, means for notifying the result of the comparison.

(10) The present invention concerns also a method for sensing an over-temperature of a power semiconductor, characterized in that the method comprises the steps of: applying a current pulse through control electrodes of the power semiconductor, duplicating the current pulse and providing the duplicated current pulse to an emulating device, comparing the voltage across the control electrodes to the voltage across the emulating device, notifying the result of the comparison.

(11) Thus, the present invention can detect an over temperature of the power semiconductor and avoid a thermal runaway without complexity. Due to the emulating device, it is not necessary to introduce any analogue to digital converter to read the actual temperature.

(12) According to a particular feature, the means for notifying the result of the comparison further control the switching pattern of the power semiconductor according to the signal outputted by the comparator.

(13) According to a particular feature, the control electrodes the current source is applied to are a gate and an emitter or a gate and a source.

- (14) According to a particular feature, the current pulse source is provided to the power semiconductor when the power semiconductor is not switching or in a transition mode.
- (15) Thus, the user can decide to stop or slow down the power semiconductor to protect the thermal runaway. Furthermore, the voltage at the control electrodes only changes according to the current pulse and the electromagnetic and other interferences do not interfere during the sensing of the over temperature of the power semiconductor.
- (16) According to the invention, the emulating device is composed of a resistor and a capacitor connected in series.
- (17) Thus, the emulating device has the same impedance as the internal gate of the power semiconductor at a given temperature, like for example a temperature just below the maximum junction temperature. The emulating device is used to create a threshold value which changes according to the current pulse source. Then the comparison between the voltage at the electrode of the power semiconductor to the voltage at the emulating device is independent of the current pulse source values.
- (18) According to the invention, the capacitor value is equal or lower than a gate capacitance value of the power semiconductor.
- (19) Thus, the present invention can prevent false alarm at the output of the comparator.
- (20) According to the invention, the resistor value is equal or lower than an internal gate resistance value of the power semiconductor when the power semiconductor is at a temperature almost equal to the maximum junction temperature of the power semiconductor.
- (21) Thus, it is possible to define the threshold temperature, which is equal or lower than the maximum junction temperature of the power semiconductor.
- (22) According to the invention, the resistor value is adjusted during a calibration phase.
- (23) Thus, it is possible to compensate the variation of the internal gate resistance of each power semiconductor.
- (24) According to the invention, the current pulse duration is equal to:
- (25)
$$t_{\max} < \frac{V_{\text{supply}} - V_{\text{th}} - R_G \cdot I}{I} \cdot C_G$$
- (26) where V_{th} stands for the threshold voltage for the gate of semiconductor, V_{supply} is the negative gate voltage that prevents the power semiconductor to turn-on or V_{supply} is the positive gate voltage that prevents the power semiconductor to turn-off, R_G is the internal gate resistance value of the power semiconductor when the power semiconductor is at a temperature almost equal to the maximum junction temperature of the power semiconductor, C_G is the gate capacitance value of the power semiconductor and I is the current pulse value.
- (27) Thus, the state of the power semiconductor is maintained the same during the pulse duration.
- (28) According to the invention, the device further comprises at least one resistor and at least one comparator, a first terminal of the at least one resistor being connected to the current copier and a second terminal of the at least one resistor being connected to the emulating device and the at least one comparator compares the voltage at the electrode of the power semiconductor to the voltage at the second terminal of the at least one resistor.
- (29) According to the invention, the method further comprises the step of adjusting the resistor value of the emulating device when the power semiconductor is heated at a predetermined temperature.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The characteristics of the invention will emerge more clearly from a reading of the following description of example embodiments, the said description being produced with reference to the accompanying drawings.

- (2) FIG. 1 represents an example of the architecture of a device for sensing the temperature of a power semiconductor device according to the invention.
- (3) FIG. 2 represents an example of a power semiconductor.
- (4) FIG. 3 represents the electric equivalent circuit of the power semiconductor according to the present invention.
- (5) FIG. 4 represents an example of an emulating device that is used for sensing the temperature of a power semiconductor according to the present invention.
- (6) FIG. 5 represents an example of a current copier that is used for sensing the temperature of a power semiconductor device according to the present invention.
- (7) FIG. 6 represents signals that are used for sensing the temperature of a power semiconductor according to the present invention.
- (8) FIG. 7 represents signals that may occur when at least one component value of the emulating device is not accurately defined.
- (9) FIG. 8 represents signals that occur when the maximum junction temperature is detected.
- (10) FIG. 9 represents an example of an algorithm that is used for adjusting the value of a resistor of the emulating power semiconductor.
- (11) FIG. 10 represents an example of the architecture of a device for sensing the temperature of a power semiconductor device according to the present invention.
- (12) FIG. 11 represents another example of the architecture of a device for sensing the temperature of a power semiconductor device according to the invention.
- (13) FIG. 12 represents an example of an algorithm for sensing the temperature of a power semiconductor device according to the invention.

DESCRIPTION OF EMBODIMENTS

- (14) FIG. 1 represents an example of the architecture of a device for sensing the temperature of a power semiconductor device according to the invention.
- (15) The device for sensing the temperature of a power semiconductor comprises a current copier **70** that duplicates the current provided by a current pulse source **60** through control electrodes of a power semiconductor **Sc 10** when the power semiconductor is not switching or in a transition mode.
- (16) Control electrodes are a gate and an emitter or a gate and a source.
- (17) The duplicated current is provided to an emulating device **20**.
- (18) The sensed voltage across the gate and the emitter or the gate and the source voltage of the power semiconductor **10** is provided to a comparator **30** that compares the voltage of the emulating device **20**.
- (19) In a variant, instead of sensing the voltage across the control electrodes to the voltage across the emulating device, the voltage is sensed across one control electrode and a reference, like for example, the ground.
- (20) If the voltage of the emulating device is upper than the sensed electrode voltage of the power semiconductor **10**, the output of the comparator **30** reaches high level.
- (21) The device for sensing the temperature of a power semiconductor comprises a controller **50** that notifies the result of the comparison by transferring a predetermined signal or a predetermined message or generates a predetermined information through a man machine interface.
- (22) In a particular mode of realization, the controller further controls the gate to source voltage of the power semiconductor and modifies the gate to source voltage patterns applied to the power semiconductor device during operation in order to reduce the junction temperature of the power semiconductor **10**.
- (23) The emulating device is an emulating impedance which is representative of the input impedance, like the one of the gate/emitter path, of the power semiconductor **10** at the desired temperature to be detected.
- (24) The pulse duration of the pulse current source **60** is controlled to certain periods, typically

several microseconds, to avoid switching-on or switching-off the power semiconductor. Then, the voltage across the real transistor is compared to the voltage across the emulating impedance. Thus, when the voltage measured is higher than the voltage at the emulating device, meaning that the gate resistance has a higher value than the emulating device one, the junction temperature of the power semiconductor **10** is hotter than the target point, the comparator **30** outputs a signal indicating the temperature overrun. Finally, the over temperature signal is fed to the controller **50** that notifies the result of the comparison.

(25) In a particular mode of realization, the controller **50** further reduces switching speed of power semiconductor or stops switching for a moment.

(26) The present invention permits to detect the over temperature during the normal operation of the power semiconductor, without any additional device near the power semiconductor, with a fast bandwidth, without any analogue to digital converter or need for digital processing.

(27) FIG. 2 represents an example of a power semiconductor.

(28) The power semiconductor is for example a transistor T and a diode D the anode of which is connected to the emitter of the transistor T and the cathode of which is connected to the collector of the transistor T.

(29) FIG. 3 represents the electric equivalent circuit of the power semiconductor according to the present invention.

(30) The electric equivalent circuit of the power semiconductor is composed of a resistor R_g connected in series with a capacitor C_g .

(31) The present invention utilises the internal gate resistance to infer the junction temperature. The internal gate resistance is created by the polysilicon layer in Field Effect Transistors which is already present in the classical structures of the transistor. In the case of typical transistors, the value of the resistance increases as the temperature increases.

(32) FIG. 4 represents an example of an emulating device that is used for sensing the temperature of a power semiconductor according to the present invention.

(33) The emulating device **20** is composed of a resistor R_e connected in series with a capacitor C_e .

(34) The resistor R_e has the same value as R_g at the maximum safe temperature operation and the capacitor C_e is for example a high-stability (such as COG) capacitor with the same value as C_g .

(35) FIG. 5 represents an example of a current copier that is used for sensing the temperature of a power semiconductor according to the present invention.

(36) The current copier **70** is for example composed of gain-matched transistor pairs.

(37) FIG. 6 represents signals that are used for sensing the temperature of a power semiconductor according to the present invention.

(38) The waveforms of the pulsed current **60**, the voltages **10o** of the power semiconductor **10** and the voltage of the emulating device **20** and the over-temperature signals **50o** are shown in FIG. 6 for three pulses.

(39) For the first pulse **60a**, the power semiconductor **10** temperature is equal to the maximum junction temperature. For the second pulse **60b**, the power semiconductor **10** temperature is inferior to the maximum junction temperature. The voltage **10ob** is upper than the voltage **20ob**. For the third pulse **60c**, the power semiconductor temperature is superior to the temperature threshold i.e. the maximum junction temperature. The voltage **20oc** is upper than the voltage **10oc** and the output signal **50o** of the comparator **30** changes of state. In the Example of FIG. 6, $C_e = C_g$ is assumed and the temperature variations are solely due to the temperature variance of the power semiconductor gate resistance.

(40) FIG. 7 represents signals that may occur when at least one component value of the emulating device is not accurately defined.

(41) According to the invention, the capacitor C_e is strictly smaller or equal to C_g , avoiding the generation of false alarms. As shown in FIG. 7, this condition is necessary and sufficient to maintain the voltage **10o** inferior to the voltage **20o** at the end of the current pulse for a resistor

value R_g inferior to the resistor value R_e . For the first pulse **60a**, the capacitor value C_e is equal to the capacitor value C_g , no false alarm is possible, even for infinite pulse duration. In the second pulse **60b**, the capacitor value C_e is superior to the capacitor value C_g , then even for a resistor value R_g smaller than the resistor value R_e , a false trigger is outputted. Finally, in the third pulse **60c**, the capacitor value C_e is lower than the capacitor value C_g , no false alarm is possible, even for infinite pulse duration.

(42) Based on the fundamental equations of the power semiconductor **10** and the emulating device voltages, where I is the current value, and t is the time:

$$(43) 10o = R_g \cdot \text{Math. } I + \frac{I}{C_g} \cdot \text{Math. } t \quad 20o = R_e \cdot \text{Math. } I + \frac{I}{C_e} \cdot \text{Math. } t$$

For R_g inferior to R_e , the voltage **10o** should be inferior to **20o** at the end of the current pulse, $t=t_{\max}$ as per. Then:

$$(44) C_e < \frac{t_{\max}}{(R_g - R_e) + \frac{1}{C_g} \cdot \text{Math. } t_{\max}} \quad \text{Where } t_{\max} < \frac{\text{Math. } V_{\text{supply}} - V_{\text{th}} \cdot \text{Math. } -R_g \cdot \text{Math. } I}{I} \cdot \text{Math. } C_g$$

where V_{th} stands for the threshold voltage for the gate of semiconductor **10**, V_{supply} is the negative gate voltage that prevents the power semiconductor to turn-on or V_{supply} is the positive gate voltage that prevents the power semiconductor to turn-off. Considering that the term, $R_{\text{sub.g}} - R_{\text{sub.e}}$ close to zero, then $C_e < C_g$.

In a particular feature, the resistor R_e is set to a slightly smaller or equal value of the resistor R_g at the maximum temperature. The value of the resistor R_e is calculated using the following formula:

$$(45) R_e = R_g^{\text{@Tmax}} + \left(\frac{1}{C_g} - \frac{1}{C_e}\right) \cdot \text{Math. } t_{\text{cmp}}$$

Where $R_{\text{sub.g.sup.@Tmax}}$ is the resistor value at maximum junction temperature.

(46) Thus, the difference between the two capacitors is compensated and an over-temperature detection becomes possible in a detection time of the comparator, noted t_{cmp} in FIG. 7.

(47) FIG. 8 represents signals that occur when the maximum junction temperature is detected.

(48) In case of the detection time t_{cmp} is too short and not easily detected, it is possible to introduce a latching circuit with set/reset function for the output of comparator.

(49) FIG. 9 represents an example of an algorithm that is used for adjusting the value of a resistor of the emulating power semiconductor.

(50) At step **S80**, the power semiconductor is heated at a predetermined temperature, for example at the maximum junction temperature minus one degree.

(51) At step **S81**, the over-temperature signal outputted by the comparator **30** is checked. If the signal is at high level, the algorithm moves to step **S81**. Otherwise, the present algorithm is interrupted.

(52) At step **S82**, the resistor value R_e value is adjusted using a laser trimming technique or fuse blowing.

(53) In a particular feature, the resistor value R_e is adjusted by an increment if the over-temperature signal outputted by the comparator is in high level. For example, the increment is equal to 1000 ppm of the estimated R_g resistor value, e.g. 5 mΩ for R_g with a nominal value of 5Ω.

(54) After that, the algorithm returns to step **S81**.

(55) FIG. 10 represents an example of the architecture of a device for sensing the temperature of a power semiconductor device according to the present invention.

(56) The device **60** has, for example, an architecture based on components connected together by a bus **901** and a processor **900** controlled by a program.

(57) The bus **901** links the processor **900** to a read only memory ROM **902**, a random access memory RAM **903**, and an input output I/O interface I/F **905**.

(58) The input output interface I/F **905** comprises the current pulse source, the current copier **70**, the emulating device **20** and the comparator **30**.

(59) The memory **903** contains registers intended to receive variables and the instructions of the program related to the algorithm disclosed in FIG. 12.

(60) The read-only memory, or possibly a flash memory **902**, contains instructions of the program

related to the algorithm disclosed in FIG. 12.

(61) When the device is powered on, the instructions stored in the memory **903** are transferred to the random access memory **903**.

(62) The device may be implemented in software by execution of a set of instructions or program by a programmable computing machine, such as a PC (Personal Computer), a DSP (Digital Signal Processor) or a microcontroller; or else implemented in hardware by a machine or a dedicated component, such as an FPGA (Field-Programmable Gate Array) or an ASIC (Application-Specific Integrated Circuit).

(63) In other words, the device includes circuitry, enabling the controller **50** to perform the program.

(64) The device **60** controls the gate signal of the semiconductor through the input output I/O interface I/F **905**, controls the switching of the power semiconductor and may control a laser trimming device for adjusting the value of the resistor of the emulating device.

(65) FIG. 11 represents another example of the architecture of a device for sensing the temperature of a power semiconductor device according to the invention.

(66) The device for sensing the temperature of a power semiconductor **10** comprises a current copier **70** that duplicates the current provided by a current pulse source **60** through control electrodes of a power semiconductor Sc **10**.

(67) The duplicated current is provided to a first terminal of a resistor R1 and to a first input of a comparator **31**.

(68) A second terminal of the resistor R1 is connected to a first terminal of a resistor R2 and to a first input of a comparator **32**.

(69) A second terminal of the resistor R2 is connected to the emulating device **20**.

(70) It has to be noted here that in the example of FIG. 11, two resistors and two comparators are added to the device disclosed in FIG. 1. Only one resistor and one comparator may be added to the device disclosed in FIG. 1 or more than two resistors and two comparators may be added to the device disclosed in FIG. 1.

(71) The sensed voltage across the control electrodes of the power semiconductor **10** is provided to the comparators **31**, **32** and **33** that compares the voltage of the emulating device **20**.

(72) If the voltage of the emulating device **20** is upper than the sensed voltage across the control electrodes of the power semiconductor **10**, the output of the comparator **30** reaches high level.

(73) The device for sensing the temperature of a power semiconductor comprises a controller **50** that notifies the result of the comparison.

(74) According to a particular feature, the controller **50** modifies the patterns applied to the power semiconductor device during operation in order to reduce the junction temperature of the power semiconductor **10**.

(75) The emulating device is an emulating impedance which is representative of the input impedance (gate/emitter path) of the power semiconductor **10** at the desired temperature to be detected.

(76) Thus, different critical temperatures of the power semiconductor **10** may be detected. The resistor Re value is adjusted for a given temperature using one of the techniques described above. The resistor R1 and R2 values are chosen in function of the temperature sensitivity of the resistor Rg value multiplied by temperature steps. Taking the numerical example above, if the temperature steps are 5° C. and the temperature sensitivity is 5 mΩ/° C., thus the resistor R1 and R2 values are 25 mΩ.

(77) The pulse duration of the pulse current source **60** is controlled to certain periods, typically several microseconds, to avoid switching-on or switching-off the power semiconductor. Then, the voltage across the electrode of the semiconductor is compared to the voltage across the emulating impedance.

(78) FIG. 12 represents an example of an algorithm for sensing the temperature of a power

semiconductor device according to the invention.

- (79) At step S110, a current pulse is applied through control electrodes of the power semiconductor.
- (80) At step S111, the current pulse is duplicated, and the duplicated current pulse is provided to the emulating device.
- (81) At step S112, the voltage across the control electrodes of the power semiconductor is compared to the voltage at the emulating device.
- (82) At step S113, the result of the comparison is notified.
- (83) The switching pattern of the power semiconductor may be also controlled according to the result of the comparison.

Claims

1. A device for sensing an over-temperature of a power semiconductor, wherein the device comprises: a current pulse source that is provided through control electrodes of the power semiconductor, a current copier that duplicates the current provided by the current pulse source, and that provides the duplicated current to an emulating device, a comparator that compares the voltage across the control electrodes to the voltage across the emulating device, and a controller configured to notify the result of the comparison, wherein the emulating device comprises a resistor and a capacitor connected in series, the resistor value is equal or lower than a gate resistance value of the power semiconductor when the power semiconductor is at a temperature equal to one degree Celsius less than the maximum junction temperature of the power semiconductor.
2. The device according to claim 1, wherein the controller is configured to control the switching pattern of the power semiconductor according to the signal outputted by the comparator by modifying a pattern applied to the power semiconductor device based on the comparator indicating that the resistance of the power semiconductor is greater than the resistance of the resistor.
3. The device according to claim 1, wherein the control electrodes the current source is applied to are a gate and an emitter or a gate and a source.
4. The device according to claim 2, wherein the current pulse source is provided to the power semiconductor when the power semiconductor is not switching or in a transition mode.
5. The device according to claim 1, wherein the capacitor value is equal or lower than a gate capacitance value of the power semiconductor.
6. The device according to claim 5, wherein the resistor value is adjusted during a calibration phase.
7. The device according to claim 6, wherein the current pulse duration is equal to:
$$t_{\max} < \frac{V_{\text{supply}} - V_{\text{th}}}{I} - R_G \cdot C_G$$
where V_{th} stands for the threshold voltage for the gate of the power semiconductor, V_{supply} is the negative gate voltage that prevents the power semiconductor to turn-on or V_{supply} is the positive gate voltage that prevents the power semiconductor to turn-off, R_G is the gate resistance value, C_G is the gate capacitance value of the power semiconductor and I is the current pulse value.
8. The device according to claim 1, wherein the device further comprises at least one other comparator, a first terminal of the resistor is connected to the current copier and a second terminal of the resistor is connected to the emulating device and the at least one other comparator compares the voltage at the electrode of the power semiconductor to the voltage at the second terminal of the at least one resistor.
9. A method for sensing an over-temperature of a power semiconductor, wherein the method comprises the steps of: applying a current pulse through control electrodes of the power semiconductor, duplicating the current pulse and providing the duplicated current pulse to an emulating device, comparing the voltage across the control electrodes to the voltage across the emulating device, notifying the result of the comparison using a controller, wherein the emulating

device comprises a resistor and a capacitor connected in series, the resistor value is equal or lower than a gate resistance value of the power semiconductor when the power semiconductor is at a temperature equal to one degree Celsius less than the maximum junction temperature of the power semiconductor.

10. The method according to claim 9, wherein the method further comprises the step of adjusting the resistor value when the power semiconductor is heated at a predetermined temperature.

11. The device according to claim 1, wherein the equivalent circuit of the power semiconductor being composed of a gate capacitor and a resistor connected in series with the gate capacitor, and the capacitor value is equal or lower than the gate capacitance value of the power semiconductor.

12. The method according to claim 9, wherein the equivalent circuit of the power semiconductor being composed of a gate capacitor and a resistor connected in series with the gate capacitor, and the capacitor value is equal or lower than the gate capacitance value of the power semiconductor.

13. The method according to claim 1, wherein the controller is configured to control the switching pattern of the power semiconductor according to the signal outputted by the comparator by reducing a switching speed of the switching pattern of the power semiconductor based on the comparator indicating that the resistance of the power semiconductor is greater than the resistance of the resistor.
